

A Bite into the Data: A Comprehensive Deep Dive into a Healthier Snacking Business

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “A Bite into the Data: A Comprehensive Deep Dive into a Healthier Snacking Business”. I extend my appreciation to Poshtique, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

A handwritten signature in black ink, appearing to read 'Henil', with a long horizontal stroke underneath.

Signature of Candidate

Name: Henil Diwan

Date: 01/03/2025

1 Executive Summary

This project focuses on the exploratory data analysis of Poshtique, a Mumbai-based food and nutrition company, primarily offering healthy alternatives to unhealthy snacks, founded by Purnima and Noopur Mehta.

The company faces stagnation of revenue and an inability to maximize profits. This is due to various challenges like understocking and overstocking items due to seasonal demand and product popularity, difficulties understanding repeat customer patterns and ineffective marketing from lack of customer insights. These result in limited customer engagement, wasted resources, missed sales opportunities and ultimately, a net loss of revenue.

Solutions to these issues could be derived from a combination of qualitative and quantitative data analysis. Qualitative insights, gathered through interviews, help understand the problem, while quantitative analysis examines sales and order data. Tools like Excel, Python and Power BI will be used to clean and model the data through visual representations, which will then inform actionable objectives to potentially mitigate the found issues.

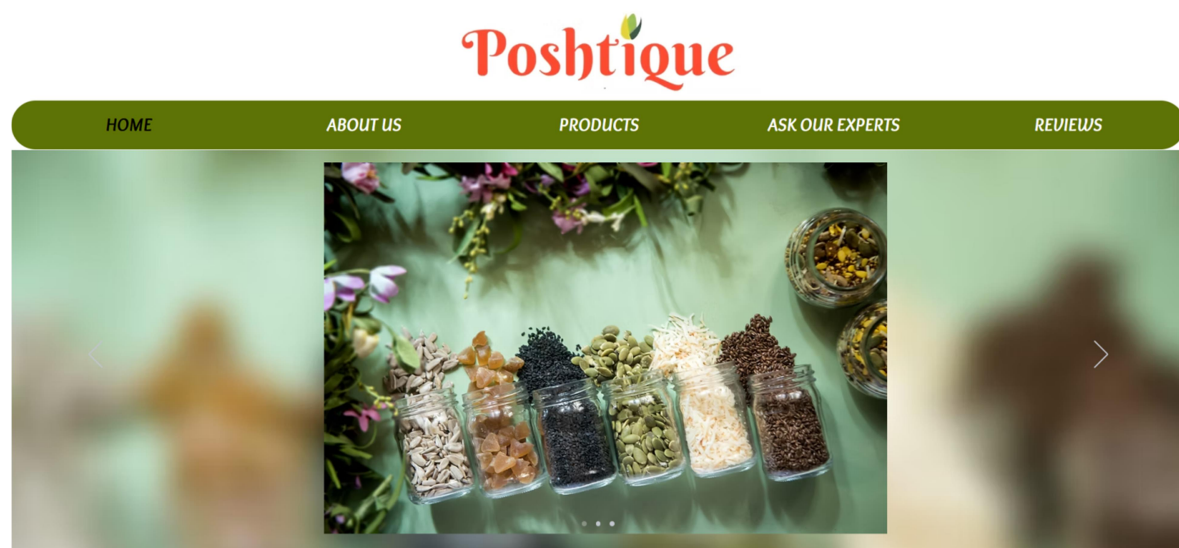
The chief outcomes of the project is to identify operational gaps and use data to harness patterns, crafting solutions to address these challenges, thus maximizing profits of the company, optimizing stock management and improving customer retention through statistical data analysis.

2 Organization Background

The company that I have collaborated with in the making of this project is 'Poshtique.' Poshtique is a Mumbai-based B2C food and nutrition brand focusing primarily on healthy alternatives to prevalent junk food items and guilt-free snacking. They target customers through a mix of two channels, Amazon and their own Online Platform.

Born from the visionary mind of Purnima Mehta and Noopur Mehta, Poshtique aims to 'Let Food be thy Medicine' through its wide array of healthy snacking options for the modern time-crunched professional, such as stuffed dates and super seeds. Hailing from a family rooted in medicinal practices, Mrs. Mehta observed the rising rate of lifestyle-related disorders, and chose

to combine it with her passion and enthusiasm of the culinary arts to create ‘Poshtique,’ with the goal of revolutionizing the snacking industry in India.



3 Problem Statement

- 3.1 Overstocking and Understocking: Poshtique often finds itself struggling with overstocking and understocking, resulting in wasted resources and missed sales opportunities.
- 3.2 Repeat Customers: Poshtique is frequently unable to understand the patterns of repeat customers due to a lack of a method for analyzing the frequency of repeat customers.
- 3.3 Marketing Strategies: Past marketing campaigns have not received the expected traction. They lack sufficient insights and data about customer distribution and preferences.

4 Background of the Problem

Effective management of inventory is a persistent challenge for many companies. Poshtique struggles with this due to seasonal fluctuations in demand, product popularity, the perishable nature of products, and rapidly changing food trends. This leads to overstocking of products that are less popular or are in a lower demand season, leading to wastage of resources and

ultimately, revenue. It also leads to understocking of products during their peak demand, missing sale opportunities.

Despite initial success and a steady growth rate in customer purchases, Poshtique faces challenges in maintaining a high rate of repeat purchases and customer loyalty. Evolving consumer preferences and customer satisfaction may influence repeat purchases. External factors such as intense local competition and price sensitivity in the vibrant Mumbai food industry could also play roles in influencing repeat purchases. A comprehensive data analysis is required to understand repeat purchasing patterns of customers.

Brand recognition plays an integral role in the growth of a company in the incredibly fragmented food industry. Developing effective market strategies poses a challenge for Poshtique, as they lack data analytics to adjust strategies and target a broader audience. Moreover, a lack of statistically verifiable data also hampers these market strategies.

5 Problem Solving Approach

Methods

A mix of qualitative and quantitative analysis will be applied to solve the above problem statements. Qualitative analysis will be conducted through interviews with the owner and senior staff. Insights from these interviews would provide a better understanding of the problem statements and the company's ground reality. These insights could also be used to supplement the findings of the quantitative analysis for added validity.

Quantitative analysis will be carried out to gather meaningful insights from the provided data. Data would first be cleaned and preprocessed to convert it into a structured format, such as removing null values and normalizing numerical data. It will then be modelled into textual and visual representations such pivot, pareto, and time-series charts to understand trends in the data, such as percentage of repeat customers, seasonal sales of types of products, total sales of products, etc. The final insights would then be used to craft solutions.

Data

The most beneficial data for gaining actionable insights would be sales data. For this specific project, data of previous sales has been provided as order summary data and order detail data. The order summary data contains customer details, date, total, tax, shipping, etc. This data could provide valuable insights into the repeat customer purchase patterns, as well as general data for targeting marketing strategies. The order details data contain specific item details for each order, such as item, quantity and stock. This data is crucial in the analysis of the stock, such as popular and seasonal items, as well as helping in the analysis of market strategies and products purchased by repeat customers. Data from previous surveys conducted by the Company would also be used to reach statistically verifiable conclusions to boost marketing campaigns.

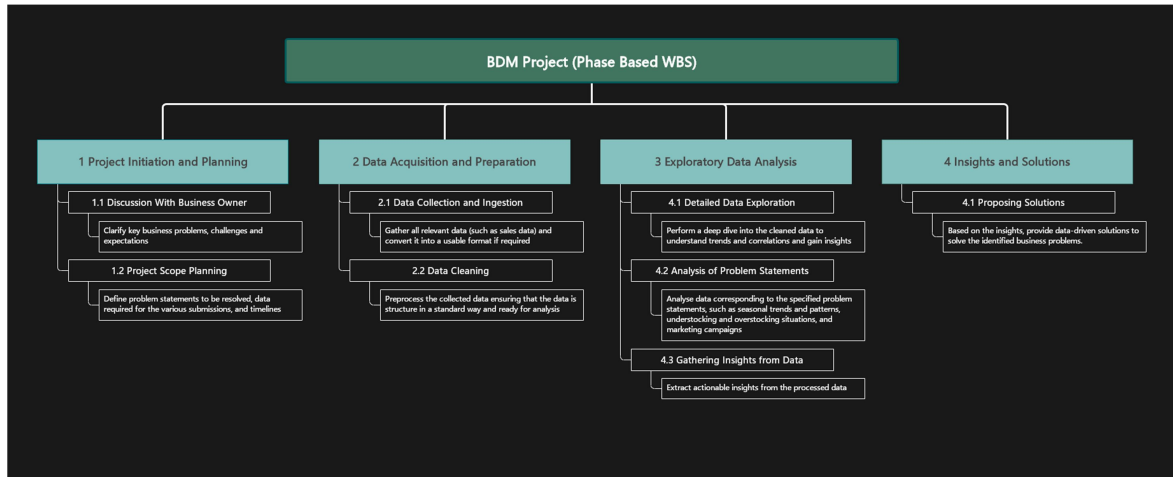
Tools

The primary tools for solving the problem statements are Microsoft Excel, Python, and Microsoft Power BI. Microsoft Excel would be used for initial data cleaning and preprocessing, as the data is provided in the form of .csv files, as well as for basic data modelling. Python would be used for more advanced data modelling and gaining insights through visual representations due to its simple syntax and extensive library selection for data science, such as NumPy, Pandas, Matplotlib, Seaborn, etc. Microsoft Power BI would be used in the creation of a highly interactive and visual dashboard to gain insights from the vast data.

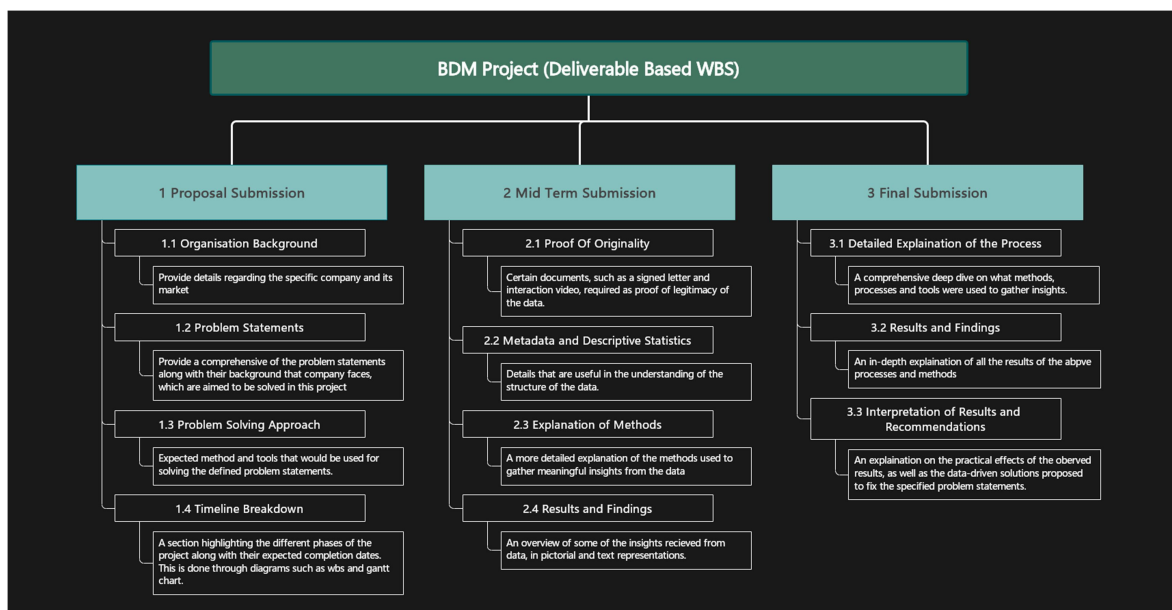
6 Expected Timeline

6.1 Work Breakdown Structure:

Phase- Based WBS -



Deliverable Based WBS –



6.2 Gantt chart



Figure - Expected timeline for completion of project.

7 Expected Outcomes

- 7.1 Improving inventory management to avoid overstocking and understocking difficulties through the use of demand forecasting and product popularity analysis.
- 7.2 Tailoring product offerings through the understanding of seasonal demands, thus maximizing profits based on season.

- 7.3 Understanding of distribution and amount of repeat customers, enhancing opportunities for customer retention, contributing to long-term stability.
- 7.4 Gaining insights into customer distribution and data, providing opportunities for targeting marketing campaigns and along with statistically verifiable data, can enhance customer engagement and loyalty.